

Numerical Integration on Real-Life Data

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About the Assignment

In this assignment I used real stock data: the daily closing price of Apple (AAPL) for the last 500 trading days (27 April 2022 to 23 April 2024). The aim was to find the total area under the closing-price curve using numerical integration.

The problem is that the data is given only for whole days, but Simpson's rules need equally spaced points. So I first used Lagrange interpolation (with 4 nearby points at a time) to get the price at the points I needed. Then I applied three methods:

- Trapezoidal Rule
- Simpson's 1/3 Rule
- Simpson's 3/8 Rule

I repeated this with different numbers of intervals ($n = 6$ up to 2000) to see how the step size (h) changes the answer. The reference value was the Trapezoidal Rule applied directly on the original data, which came out as 82525.32.

Graphs



Figure 1: AAPL closing price vs trading day

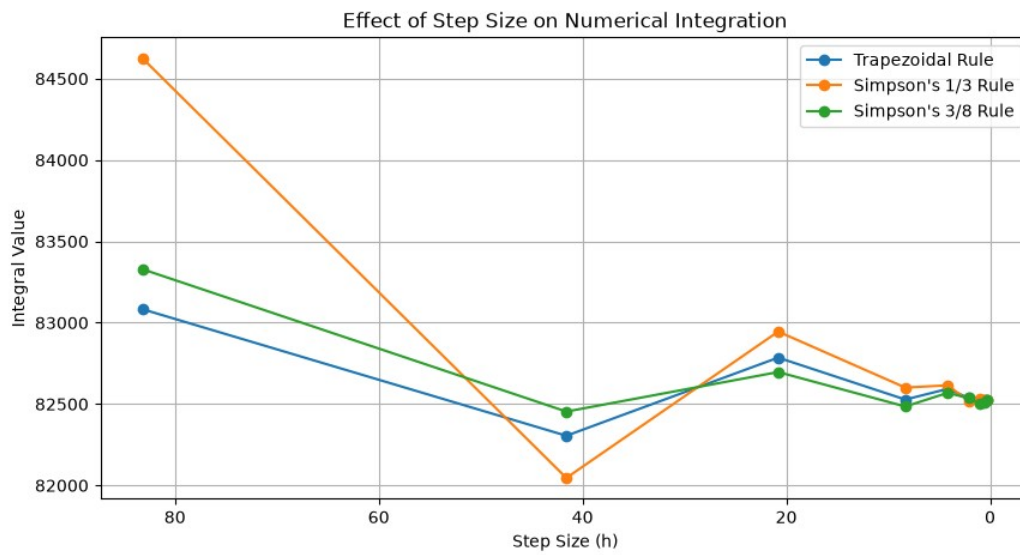


Figure 2: Integral value vs step size

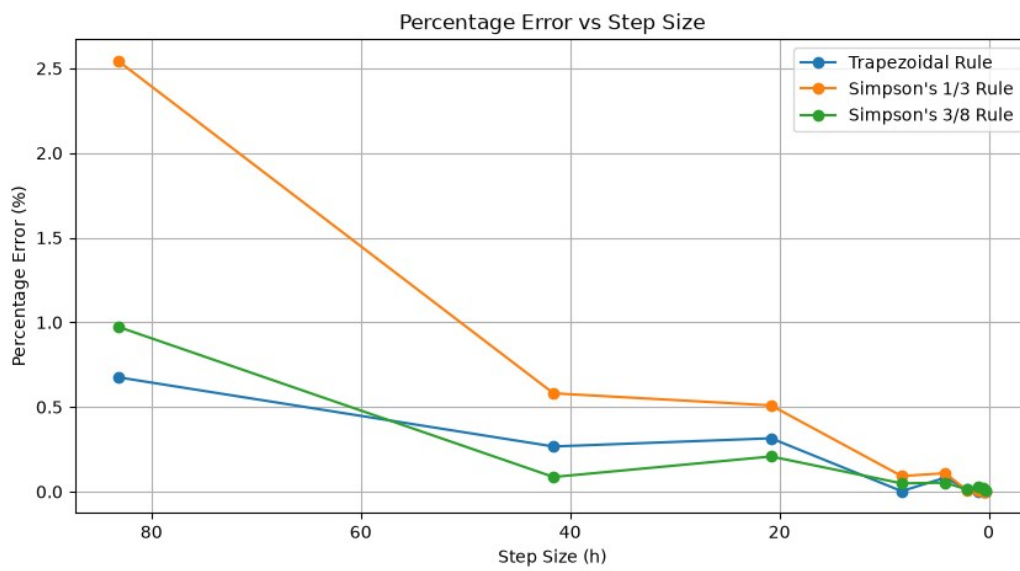


Figure 3: Percentage error vs step size

Results

Integral values for different numbers of intervals (n):

n	h	Trapezoidal	Simpson 1/3	Simpson 3/8
6	83.1667	83082.61	84624.02	83328.51
12	41.5833	82305.18	82046.04	82453.89
24	20.7917	82785.47	82945.56	82696.72
60	8.3167	82527.68	82601.24	82484.95
120	4.1583	82593.27	82615.14	82568.95
240	2.0792	82537.20	82518.51	82535.85
500	0.998	82525.34	82529.70	82501.13
1000	0.499	82526.79	82527.28	82506.59
2000	0.2495	82527.16	82527.28	82522.20

Percentage error compared with the reference value:

n	Trapezoidal	Simpson 1/3	Simpson 3/8
6	0.675%	2.543%	0.973%
12	0.267%	0.581%	0.087%
24	0.315%	0.509%	0.208%
60	0.003%	0.092%	0.049%
120	0.082%	0.109%	0.053%
240	0.014%	0.008%	0.013%
500	0.00002%	0.005%	0.029%
1000	0.002%	0.002%	0.023%
2000	0.002%	0.002%	0.004%

At $n = 60$, the Trapezoidal Rule had an error of only about 0.003%. At $n = 2000$, all three methods were within about 0.004% of the reference value.

Conclusion

All three methods gave answers very close to the reference value of about 82525. When the step size was made smaller, the error mostly went down, and it became very small for large n .

The error did not go down smoothly every time. For example, $n = 60$ was more accurate than $n = 120$ for the Trapezoidal Rule. This is because stock prices go up and down randomly, unlike a smooth curve.

Overall, this assignment showed that interpolation lets us use numerical integration even when we only have data at separate points, like real stock prices.